

Updated Project Pipeline

Supplier Lead-Time Risk Prediction — DAP391m, Summer 2026

Group 8

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Document Purpose

This document records every pipeline decision made after receiving supervisor feedback (“data not good enough”) and reviewing the reference paper by Orajaka & Okolie (2025). It specifies exactly what to adopt, what to keep, what to drop, and how each pipeline script must change. No code is modified by this document; it is the specification that guides code changes.

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1 Executive Summary

The supervisor’s assessment that the original data was “ok but not good enough” was confirmed by quantitative analysis: the original joined dataset produced only 704 rows from 7 unique suppliers, providing insufficient coverage for reliable generalization. The team sourced a replacement dataset (`supply_chain_risk_dataset.csv`) containing 2,478 rows across 51 suppliers, which directly resolves the scale concern.

In parallel, the reference paper—*Using Artificial Intelligence to predict and optimize supplier lead times in procurement operations* (Orajaka & Okolie, 2025)—was reviewed in full. While the paper adopts a regression framing that ultimately fails ($R^2 = -0.09$), it contributes three high-value techniques: expanding-window supplier history features, temporal order features, and a rigorous theoretical framework spanning five management theories.

The central update is therefore two-fold: migrate to the new dataset and integrate the paper’s best feature-engineering practices. The project’s classification framing, four-model benchmark, PR-AUC primary metric, and stratified cross-validation are all retained—they are demonstrably stronger than the paper’s approach and better suited to the procurement risk-alerting use case.

2 Dataset Transition

2.1 Quantitative Comparison

Table 1 presents a side-by-side comparison of the original three-CSV dataset and the new single-CSV dataset across the dimensions most relevant to model reliability.

Table 1: Original vs. new dataset — key metrics		
Dimension	Original (3 CSVs)	New Dataset
Rows after join	704	2,478
Unique suppliers	7	51
Missing values	Present	Zero
Target variable	<code>delivery_status</code> (binary)	<code>risk_label</code> (3-class)
Class distribution	16% Delayed / 84% On-Time	63% High / 19% Low / 18% Medium
Supplier features	<code>lead_time_days</code> , <code>satisfaction_score</code>	<code>supplier_lead_time_days</code> , <code>supplier_quality_score</code> , <code>supplier_reliability_index</code>
Environmental features	None	Port delay, weather disruption, fuel price, market demand
Machine/IoT features	None	Temperature, vibration, run- time
Temporal coverage	Multiple years	Mar–Jul 2025 (5 months)

2.2 Target Variable Decision

The new dataset provides two target columns: `risk_probability` (continuous 0–1) and `risk_label` (categorical: High/Medium/Low). A correlation analysis revealed that `risk_probability` correlates at $r = 0.898$ with `port_delay_days`, confirming the label is synthetically derived rather than observed from real delivery outcomes. The detailed implications are addressed in Section 3.

For modelling, the project will use `risk_label` as the primary target in its three-class form (High, Medium, Low). A binary collapse (High vs. not-High) will be computed as a secondary comparison target to maintain continuity with the original proposal’s binary framing.

3 Data Quality Warning: Leakage Risk

Critical Warning

Finding: `port_delay_days` correlates with `risk_probability` at $r = 0.898$. All other features correlate at $r \leq 0.102$. The target label is therefore an algebraic function of `port_delay_days`, not an independent business outcome.

This finding has two direct pipeline consequences. First, `port_delay_days` must be excluded from the model feature set entirely. A model trained with it as a feature would learn a trivial mapping and achieve near-perfect accuracy on a proxy signal rather than a real procurement risk. Second, `risk_probability` must also be excluded for the same reason—it *is* the target transformed.

The remaining 12 operational and environmental features (temperature, vibration, runtime, inventory level, pending orders, supplier lead time, supplier quality score, supplier reliability index, fuel price index, market demand index, and weather disruption score) represent genuine supply chain signals. After adding the engineered features described in Section 4, the final feature matrix will contain approximately 17–20 columns.

The synthetic nature of the target must be disclosed in the project report’s data section. A suggested sentence: “The *risk_label* target variable is derived primarily from port delay conditions ($r = 0.898$), making it a proxy risk score rather than a directly observed delivery outcome. To prevent target leakage, `port_delay_days` is excluded from the feature set.”

4 Updated Feature Engineering Plan

4.1 Features Adopted from the Paper

Expanding-Window Supplier History. The paper’s second and third most important SHAP features are `supplier_avg_lead` and `supplier_std_lead`, computed using an expanding window per supplier. These capture a supplier’s historical reliability: the mean quantifies average performance, while the standard deviation captures consistency. Both will be engineered from `supplier_lead_time_days` in the new dataset. A critical implementation detail is the `.shift(1)` offset, which ensures the current row’s lead-time value does not contribute to its own prediction feature, thereby preventing look-ahead leakage.

Expanding Window — Implementation Spec

```
df = df.sort_values('timestamp')
df['supplier_avg_lead'] = df.groupby('supplier_id')['supplier_lead_time_days'] \
    .transform(lambda x: x.expanding().mean().shift(1))
df['supplier_std_lead'] = df.groupby('supplier_id')['supplier_lead_time_days'] \
    .transform(lambda x: x.expanding().std().shift(1))
```

With ≈ 48.6 observations per supplier on average, the expanding windows will have sufficient history to stabilise by the 10th observation per supplier.

Temporal Features. The paper extracts `order_month` and `order_weekday` from the order date to capture seasonal and scheduling patterns. The new dataset’s `timestamp` column enables

equivalent extraction plus an `order_hour` feature that reflects shift-level operational patterns—a dimension the paper did not exploit.

4.2 Original Project Features to Engineer

Supplier Risk Score. Combining multiple supplier signals into a single composite score makes the signal denser and provides an interpretable KPI for the Power BI dashboard. The composite weights supplier lead-time history (40%), lead-time consistency (30%), quality score (20%), and reliability index (10%). These weights are heuristic starting points; SHAP analysis after modelling will validate whether the composite or its constituent parts better explain predictions.

Supplier Risk Score — Implementation Spec

```
df['supplier_risk_score'] = (
    df['supplier_avg_lead'] * 0.40 +
    df['supplier_std_lead'] * 0.30 +
    (100 - df['supplier_quality_score']) / 100 * 0.20 +
    (1 - df['supplier_reliability_index']) * 0.10
)
```

Rows where `supplier_avg_lead` is NaN (first observation per supplier) should have `supplier_risk_score` set to NaN and subsequently imputed with the supplier-level median.

External Risk Composite. The paper explicitly identifies environmental factors as a gap in the literature. The new dataset fills this gap with weather disruption, fuel price, and market demand. These will be combined into a single external disruption index (after `port_delay_days` is excluded). This composite provides an interpretable environmental-pressure signal for the Streamlit app.

4.3 Encoding Changes

The original pipeline used `LabelEncoder` for `supplier_id`. This will be replaced with One-Hot Encoding for `supplier_id` (51 unique values = 50 binary columns, manageable). `machine_id` has 201 unique values; OHE would produce 200 noisy indicator columns. Instead, `machine_id` will be dropped entirely—machine-level identity is not a procurement variable and the supplier features already capture the relevant supplier-level variation.

4.4 Feature Inventory: Final Matrix

Table 2 summarises the full feature set after engineering and encoding.

5 Pipeline Script Specifications

Each of the six pipeline scripts requires updates for the new dataset. This section specifies exactly what changes in each file. No code is modified here; these are specifications for the developer to implement.

5.1 Script 01 — Ingestion and Cleaning

The script must be updated to load `Data/supply_chain_risk_dataset.csv` instead of the three original CSVs. The three-way join logic (customer → shipment → logistics) is replaced by single-file ingestion. Type coercion should parse `timestamp` as `datetime64`, and `risk_label` as a categorical with ordered levels (Low < Medium < High). The null check will pass trivially since the new dataset has zero missing values, but the check should remain as a pipeline guard. Output: `Data/filtered/clean_data.csv`.

Table 2: Final feature matrix — new dataset after engineering

Feature	Source	Notes
temperature_C	Raw	Machine operational signal
vibration_level	Raw	Machine operational signal
machine_runtime_hours	Raw	Machine utilisation
inventory_level_units	Raw	Inventory pressure
pending_orders	Raw	Demand pressure
supplier_lead_time_days	Raw	Core supplier metric
supplier_quality_score	Raw	Proxy for defect rate
supplier_reliability_index	Raw	Delivery consistency
fuel_price_index	Raw	External cost pressure
weather_disruption_score	Raw	Environmental disruption
market_demand_index	Raw	Demand-side pressure
order_month	Engineered	Seasonal pattern
order_weekday	Engineered	Scheduling pattern
order_hour	Engineered	Shift-level pattern
supplier_avg_lead	Engineered	Historical mean (paper)
supplier_std_lead	Engineered	Historical variability (paper)
supplier_risk_score	Engineered	Composite (new)
external_risk_score	Engineered	Environmental composite
supplier_id_* (OHE)	Encoded	50 binary columns
port_delay_days	DROPPED	Near-target leakage ($r = 0.898$)
risk_probability	DROPPED	Direct target leakage
machine_id	DROPPED	201 categories, no procurement signal
timestamp	DROPPED	Replaced by temporal features

5.2 Script 02 — SQL Analysis

The SQL queries must be rewritten for the new schema. The six required queries map to the new columns as follows: average lead time per supplier uses `supplier_lead_time_days` grouped by `supplier_id`; delay frequency uses `risk_label = 'High'` as the delay proxy; volume-delay correlation compares `pending_orders` against `risk_label`; monthly trends group by `strftime('%Y-%m', timestamp)`; year-over-year growth uses the same timestamp grouping at annual resolution; and the penetration index computes the fraction of High-risk events per supplier relative to the dataset mean.

5.3 Script 03 — EDA

The EDA script should add three new analyses: a correlation heatmap confirming the `port_delay_days` leakage signal (to document it explicitly), a class balance chart for the three-class target, and a per-supplier box plot of `supplier_lead_time_days` to visualise the 51-supplier spread (analogous to Figure 2 in the reference paper).

5.4 Script 04 — Feature Engineering

This is the most substantially changed script. The changes are:

1. Parse timestamp and extract `order_month`, `order_weekday`, `order_hour`.
2. Drop `port_delay_days`, `risk_probability`, `machine_id`, `timestamp`.
3. Sort by `timestamp` and compute expanding-window supplier features with shift.
4. Compute `supplier_risk_score` and `external_risk_score`.
5. Apply IQR capping to continuous features (retain existing IQR logic).
6. Replace `LabelEncoder` with `pd.get_dummies` for `supplier_id`.
7. Encode `risk_label` as ordered integer (Low=0, Medium=1, High=2) for multi-class models.
8. Stratified train/test split (80/20) preserving class proportions.

5.5 Script 05 — Modelling

The four classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost) are retained without change to their architecture. The target changes from binary to three-class, which requires two configuration updates: `multi_class='ovr'` for Logistic Regression and `objective='multi:softprob'` with `num_class=3` for XGBoost. The stratified five-fold cross-validation configuration is unchanged. Primary evaluation metrics shift from binary PR-AUC to macro-averaged PR-AUC across the three classes, with per-class recall reported separately. A binary secondary analysis (High vs. not-High) should be run alongside the three-class models to maintain comparability with the original proposal.

5.6 Script 06 — Advanced Visualisation

The Folium risk heatmap should be adapted to use `supplier_id` as the grouping key. Since the new dataset lacks geographic coordinates, a simulated spatial layout (supplier IDs mapped to a grid) or a bubble chart by risk level provides an equivalent visual. The existing Plotly boxplots by supplier and lead-time histograms translate directly to the new schema.

6 Model Evaluation Framework

6.1 Primary Metrics

Macro-averaged PR-AUC is the primary metric for the three-class setting, ensuring that performance on the minority classes (Medium and Low) is not obscured by the dominant High class.

Per-class recall is reported separately so that the High-risk recall—the most operationally critical figure—can be evaluated in isolation. This mirrors the original proposal’s emphasis on recall for the delayed class.

6.2 Secondary Metrics

ROC-AUC (macro OvR), weighted F1-score, and the three-class confusion matrix serve as secondary metrics. For the binary secondary analysis, the full set of original secondary metrics (binary PR-AUC, binary ROC-AUC, F1, confusion matrix) is preserved.

6.3 Validation

Stratified five-fold cross-validation is retained. With 2,478 rows, each fold contains approximately 496 samples, which provides stable variance estimates across all three classes. The final test set (held-out 20%) is used only for final reporting; all hyperparameter decisions are made on the cross-validation folds.

Why Not Switch to Regression?

The reference paper frames the task as regression (predicting exact lead-time days) and achieves $R^2 = -0.09$, meaning the model performs worse than a simple mean predictor. Procurement teams need an actionable risk flag (High / Medium / Low), not a day-count estimate. The classification framing is retained for both practical and performance reasons.

7 Theoretical Framework for the Report

The reference paper grounds its work in five management theories. All five directly justify the AI-driven classification approach used in this project and should be cited in the final report’s literature review section. Table 3 maps each theory to its specific justification for this project.

Table 3: Theoretical framework adopted from Orajaka & Okolie (2025)

Theory	Justification for this Project
Transaction Cost Economics (Williamson, 1979)	Lead-time delays are hidden transaction costs. ML quantifies this cost and enables procurement teams to price supplier risk into sourcing decisions.
Resource-Based View (Barney, 1991)	Supplier reliability is a strategic, firm-specific resource. Predictive modelling identifies which suppliers constitute a competitive advantage and which are a liability.
Lean Supply Chain Theory (Womack & Jones, 1996)	Delay prediction enables proactive waste elimination. By anticipating late deliveries, procurement can adjust safety-stock policies and avoid production stoppages.
Complex Adaptive Systems (Choi et al., 2001)	Supply chains are non-linear, emergent networks. Tree-based ensemble models (RF, XGBoost) are better suited to this non-linearity than linear regression.
Decision Theory + Risk Management (Raiffa, 1968)	Probabilistic risk scores from the classifier support rational procurement decisions under uncertainty, enabling cost–risk trade-off analysis.

8 Full Adoption Decision Table

Table 4 is the master decision record for this pipeline update, covering every element evaluated across the original dataset, the new dataset, and the reference paper.

Table 4: Master adoption table — pipeline update decisions

Element	Action	Rationale
supplier_avg_lead via expanding window	ADOPT	Paper's 2nd SHAP feature; captures supplier history
supplier_std_lead via expanding window	ADOPT	Paper's 3rd SHAP feature; captures delivery consistency
One-Hot Encoding for supplier_id	ADOPT	Avoids false ordinal relationships in tree models
order_month, order_weekday, order_hour	ADOPT	Temporal features from paper; new dataset enables order_hour
Five theoretical frameworks in literature review	ADOPT	Directly justifies ML approach; paper provides full citations
IQR outlier capping on continuous features	RETAIN (extend)	Already in pipeline; extend to new dataset columns
Classification framing (not regression)	KEEP	Paper's regression failed ($R^2 = -0.09$); classification is more actionable
Four-model benchmarking	KEEP	More rigorous than paper's single-model approach
PR-AUC + Recall as primary metrics	KEEP	Correct for imbalanced multi-class; paper uses inappropriate MAE/RMSE
Stratified 5-fold cross-validation	KEEP	Paper has no CV; provides variance estimates
Per-prediction SHAP waterfall in Streamlit	KEEP	Paper shows aggregate SHAP only; per-prediction is the differentiator
Power BI supplier scorecard	KEEP	Course requirement; not present in paper
SQL 6-query analysis	KEEP	Course requirement; queries rewritten for new schema
3-class target (High/Medium/Low)	ADD	More nuanced than binary; aligns with Power BI risk tiers
supplier_risk_score composite	ADD	Paper-inspired; provides interpretable KPI for dashboard
external_risk_score composite	ADD	Addresses paper's stated future-work gap
Binary secondary analysis (High vs. not-High)	ADD	Maintains comparability with original proposal
Data leakage disclosure in report	ADD	$r = 0.898$ finding must be documented
port_delay_days as model feature	DROP	Near-target leakage ($r = 0.898$)
risk_probability as model feature	DROP	Direct target leakage

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Table 4 continued

Element	Action	Rationale
machine_id OHE	DROP	201 categories; no procurement signal
Regression task (MAE / RMSE / R^2)	REJECT	Paper demonstrated failure; wrong framing
Original 3-CSV join pipeline	RETIRE	Replaced by new dataset; 7 suppliers insufficient

9 Deliverables Status

9.1 Course Requirements

All course deliverables are unaffected by the dataset change. The four-model comparison, SQL analysis file (`sql/analysis.sql`), AI Audit Log (`docs/AI_AuditLog_Template_DAP391m.xlsx`), and final LaTeX report (`report/main.tex`) remain as specified. The modelling target changes from binary to three-class, which updates the classifier configurations and evaluation tables but does not change the deliverable categories.

9.2 Applied Deliverables

The Power BI supplier scorecard gains a natural upgrade path: the three-class risk label (High/Medium/Low) maps directly to traffic-light risk alerts without requiring threshold tuning. The Streamlit application's SHAP waterfall explanation is unchanged; the input fields will be updated to match the new feature set.

9.3 Report Literature Review

A minimum of three citations from Orajaka & Okolie (2025) should be incorporated into the final report: the feature importance finding that supplier reliability dominates lead-time prediction, the gap identification for environmental factors (now addressed by the new dataset), and the recommendation for ensemble over linear models. These citations strengthen the report's positioning relative to prior work.

10 Implementation Order

The pipeline updates should be implemented in the following sequence to minimise integration risk. Scripts 01 and 04 carry the highest change surface; scripts 05 and 06 require only configuration updates once the feature matrix is validated.

1. **Script 01** — Update ingestion for new CSV; validate shape ($2,478 \times 17$) and zero nulls.
2. **Script 03** — Run EDA on raw new data; confirm leakage correlation visually.
3. **Script 04** — Implement all feature engineering changes; validate final matrix shape.
4. **Script 02** — Rewrite SQL queries for new schema; validate output tables.
5. **Script 05** — Update classifier configs for three-class; run CV; report metrics.
6. **Script 06** — Update visualisations for new schema; generate report figures.
7. **Streamlit app** — Update input fields and SHAP waterfall for new feature set.
8. **Power BI** — Rebuild scorecard with three-class risk tiers.

Running the pipeline in this order ensures that each downstream script receives a validated input before it is updated, making integration errors traceable to a single script at a time.